

# ICT-2107: Signals and Systems

## Class 1 Lecture Note

### Course Overview and Introduction to Signals and Systems

#### Basic Input-Output Model of a System

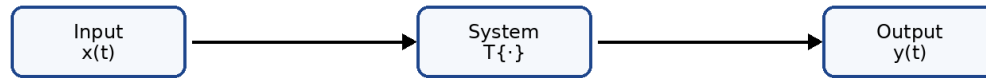


Figure 1. A system receives an input signal, processes it, and produces an output signal.

Study Material Prepared by

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Class Duration: 1 Hour

Target Students: B.Sc. Engineering, 2nd Year

Course Type: Theory

## 1. Class Objectives

- Understand the purpose and importance of the Signals and Systems course.
- Define signal and identify real-life examples of signals.
- Define system and explain the input-output model of a system.
- Distinguish between a physical phenomenon, a signal, and a system.
- Relate signals and systems concepts to ICT, communication, biomedical, image, audio, and control applications.
- Use simple mathematical notation such as  $x(t)$ ,  $x[n]$ , and  $y(t)$ .

## 2. Suggested Time Distribution

| Time   | Activity                                      |
|--------|-----------------------------------------------|
| 5 min  | Greeting, course introduction, and motivation |
| 10 min | Why study Signals and Systems?                |
| 10 min | Definition and real examples of signals       |
| 10 min | Definition and real examples of systems       |
| 10 min | Input-output model and block diagrams         |
| 10 min | Engineering applications with examples        |
| 5 min  | Summary, quick quiz, and homework             |

### 3. Board Writing Plan

Write the following on the board at the beginning of class:

```
ICT-2107: Signals and Systems  
Class 1: Course Overview and Introduction  
Prepared by: Prof. Dr. Md Zahidul Islam  
Department of ICT, Islamic University, Kushtia
```

Today's key questions:

1. What is a signal?
2. What is a system?
3. Why do ICT students study Signals and Systems?
4. How does a system process an input signal?
5. Where are signals and systems used in real life?

### 4. Course Overview

Signals and Systems is a foundation course for communication engineering, digital signal processing, control systems, image processing, audio processing, biomedical signal analysis, and many ICT applications. The main idea of the course is simple: information is usually carried by signals, and signals are processed by systems.

A mobile phone signal, speech signal, ECG signal, image, video, temperature record, internet data stream, and sensor output are all signals. A microphone, speaker, camera, amplifier, communication channel, filter, computer processor, and human ear can be modeled as systems.

### 5. What is a Signal?

A signal is a function that carries information about the behavior or nature of a physical phenomenon. In continuous time, a signal is often written as  $x(t)$ , where  $t$  represents time. In discrete time, a signal is written as  $x[n]$ , where  $n$  is an integer index.

```
Continuous-time signal:  $x(t)$   
Discrete-time signal:  $x[n]$ 
```

Simple explanation: a signal is a changing quantity that carries information. For example, when a person speaks, air pressure changes with time. That changing air pressure is a speech signal.

### 6. Real Signal Figures

The following figures are plotted signal shapes. These figures should be shown to students so that they can visualize how signals vary with time.

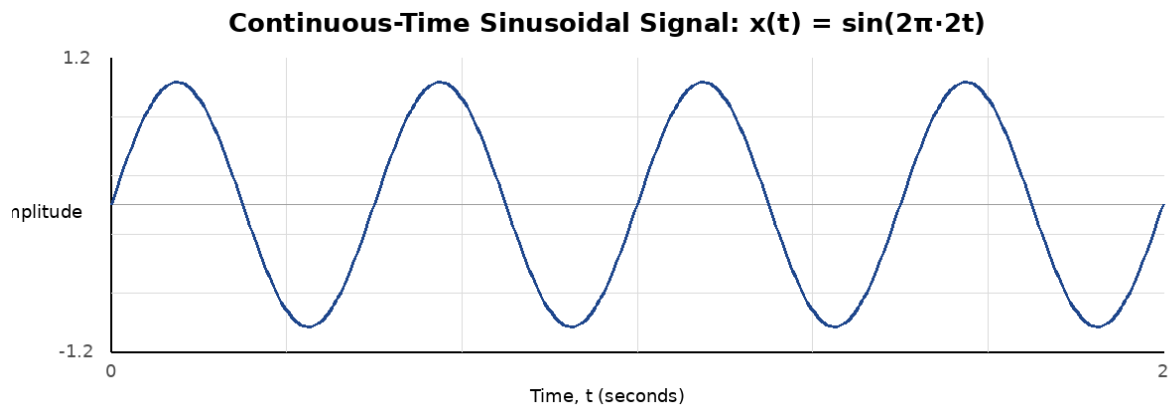


Figure 2. Sinusoidal signal. This shape appears in AC voltage, sound waves, and communication systems.

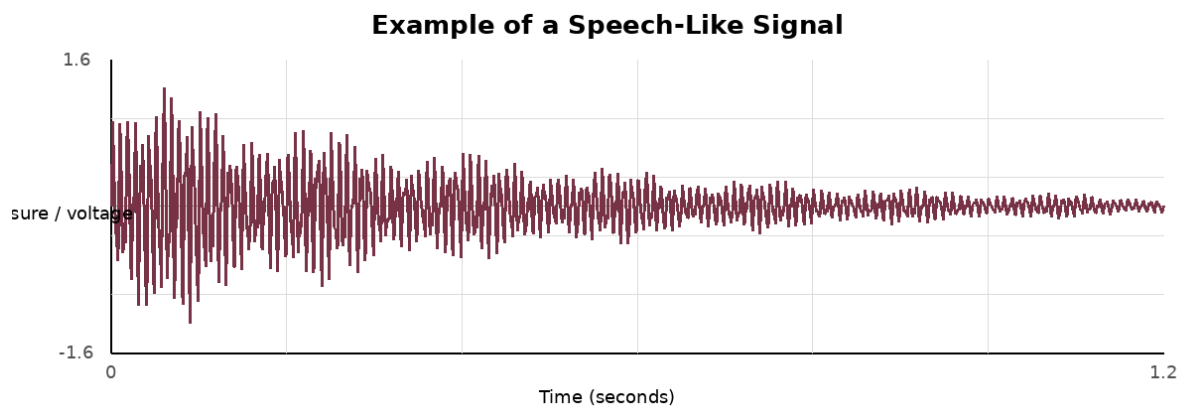


Figure 3. Speech-like signal. Human speech changes rapidly with time and carries voice information.

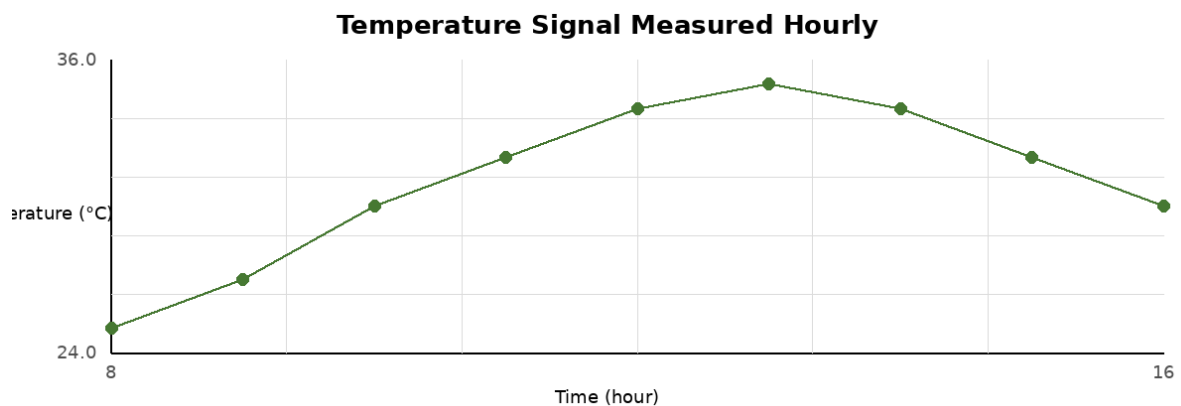


Figure 4. Temperature signal measured every hour. This is a slowly varying signal.

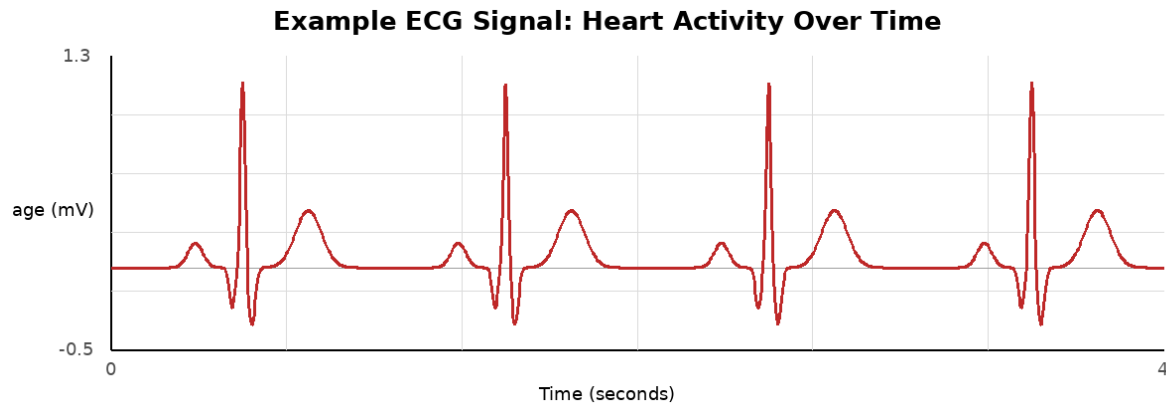


Figure 5. ECG-like biomedical signal. Heart activity can be represented as voltage variation over time.

## 7. Real-Life Examples of Signals

| Signal               | Independent Variable | Information Carried              |
|----------------------|----------------------|----------------------------------|
| Speech signal        | Time                 | Words, tone, speaker information |
| ECG signal           | Time                 | Heart electrical activity        |
| Temperature signal   | Time                 | Temperature variation            |
| Image signal         | Space: x and y       | Brightness or color information  |
| Video signal         | Time and space       | Moving visual information        |
| Internet data signal | Time                 | Digital data                     |
| Radio signal         | Time                 | Audio or message information     |

## 8. Signal as a Mathematical Function

A signal can be represented mathematically. For example:

$$x(t) = 5 \sin(2\pi t)$$

Here,  $x(t)$  is the signal value at time  $t$ . The value changes as  $t$  changes. The maximum value is called amplitude. The repetition rate is related to frequency.

## 9. What is a System?

A system is a physical device, process, algorithm, or mathematical model that takes an input signal, performs an operation on it, and produces an output signal. If the input is  $x(t)$  and the output is  $y(t)$ , the system relation is written as:

$$x(t) \rightarrow \text{System} \rightarrow y(t)$$

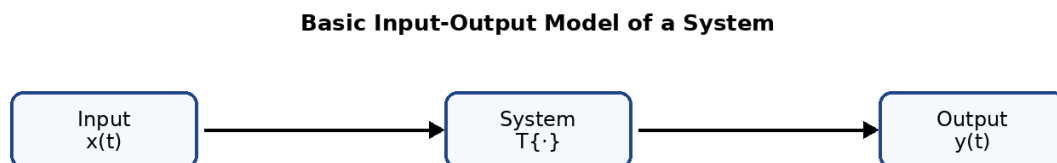


Figure 6. Basic system model used throughout Signals and Systems.

## 10. Real-Life Examples of Systems

| System      | Input Signal           | Output Signal            |
|-------------|------------------------|--------------------------|
| Microphone  | Sound wave             | Electrical signal        |
| Speaker     | Electrical signal      | Sound wave               |
| Camera      | Light/image            | Digital image            |
| Amplifier   | Weak electrical signal | Strong electrical signal |
| Filter      | Noisy signal           | Cleaner signal           |
| Human ear   | Sound wave             | Nerve signal             |
| Thermometer | Temperature            | Temperature reading      |

## 11. Example: Microphone as a System

A microphone receives sound pressure variation and converts it into an electrical voltage signal. Therefore, a microphone is a system because it transforms one signal into another signal.

Sound signal → Microphone → Electrical signal

## 12. Example: Mobile Communication System

Mobile communication is one of the best examples of signals and systems. A speaker's voice is converted into an electrical signal, processed, transmitted through a wireless channel, received by another phone, and converted back into sound.

**Mobile Communication as a Signals-and-Systems Example**

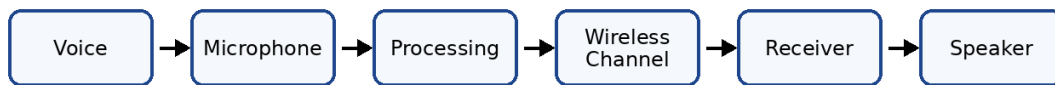


Figure 7. Mobile communication system as a chain of signal-processing systems.

## 13. Example: Filter System for Noise Reduction

A filter is a system that changes a signal. One important use of a filter is to reduce unwanted noise. For example, in a mobile phone call, background noise may be reduced so that speech becomes clearer.

**Noise Reduction Using a Filter System**



Figure 8. A filter system can reduce noise from a signal.

## 14. Simple Mathematical System Example

Suppose a system doubles the amplitude of the input signal. The system can be written as:

$$y(t) = 2x(t)$$

If the input value at a certain time is  $x(t) = 3$ , then the output is:

$$y(t) = 2 \times 3 = 6$$

This is similar to an amplifier. An amplifier increases the strength of a weak signal.

## 15. Why Signals and Systems Are Important for ICT Students

- Understanding mobile and wireless communication
- Analyzing audio, image, and video data
- Designing filters and removing noise
- Understanding digital signal processing
- Working with biomedical signals such as ECG and EEG
- Understanding sensors, IoT devices, and data acquisition
- Modeling control systems and communication channels
- Preparing for advanced courses in DSP, communication, control, and image processing

## 16. Important Terms

| Term                   | Meaning                                                                                |
|------------------------|----------------------------------------------------------------------------------------|
| Signal                 | A function that carries information                                                    |
| System                 | A device, process, algorithm, or model that transforms input signal into output signal |
| Input signal           | Signal applied to a system                                                             |
| Output signal          | Signal produced by a system                                                            |
| Continuous-time signal | Signal defined for every value of time                                                 |
| Discrete-time signal   | Signal defined only at specific time instants                                          |
| Signal processing      | Operations performed on signals to modify, analyze, or extract information             |

## 17. In-Class Discussion Questions

6. Is human voice a signal? Explain why.
7. Is a mobile phone a system? Explain why.
8. Is a photograph a signal? What are its independent variables?
9. What are the input and output of a speaker?
10. What are the input and output of a thermometer?
11. Give one signal and one system from your daily life.

## 18. Short Class Quiz

12. Define signal.
13. Define system.

14. Write the general input-output model of a system.
15. Give two examples of signals.
16. Give two examples of systems.
17. What is the input of a microphone?
18. What is the output of a speaker?
19. Why is a mobile phone considered a system?
20. What is the meaning of  $x(t)$ ?
21. What is the meaning of  $y(t)$ ?

## 19. Board Summary

Class 1 Summary:

1. A signal carries information.
2. A system processes an input signal and produces output.
3. Signals are often written as  $x(t)$  or  $x[n]$ .
4. Output signals are often written as  $y(t)$  or  $y[n]$ .
5. Examples of signals: speech, ECG, image, video, temperature.
6. Examples of systems: microphone, speaker, amplifier, filter, mobile phone.
7. Signals and Systems is important in communication, biomedical engineering, image processing, audio processing, and control systems.

## 20. Homework

22. Write short definitions of signal, system, input signal, output signal, and signal processing.
23. Give five real-life examples of signals and explain what information each signal carries.
24. Give five real-life examples of systems and identify their input and output signals.
25. Draw the input-output block diagrams of a microphone, speaker, camera, mobile phone, and audio amplifier.
26. Explain why Signals and Systems is important for ICT students.

## 21. Teacher's Closing Statement

Today we started the course Signals and Systems. We learned that a signal carries information and a system processes signals. In real life, speech, images, temperature, ECG, and videos are signals, while microphones, speakers, filters, amplifiers, cameras, and mobile phones are systems. From the next class, we will study the classification of signals, especially continuous-time, discrete-time, analog, and digital signals.